

Lecture Outline

- Visualizing Newton
- Optimization with Newton
- **Convergence of Newton's method**

Last Class

Last class, we introduced Newton's method for finding roots ($f(r)$) for a scalar function $f: \mathbb{R} \rightarrow \mathbb{R}$ from some initial guess x_0 :

$$x_{k+1} = x_k - \frac{f(x_0)}{f'(x_0)}.$$

But we have some lingering questions:

(1) How do I choose x_0 ?

(2) Does the choice of x_0 effects the root if converges?

Order of Convergence

Let x_k be the k th iterate of a root finding method with associated absolute error $e_k = r - x_k$, where r is a root of f . We say that this root finding method has an **order of convergence of d** if

$$e_k = r - x_k$$

$$|e_{k+1}| \leq C \cdot |e_k|^d \rightarrow \text{Order of convergence } d$$

for some $C \in \mathbb{R}$

$d=1$: Method converges linearly

$d=2$: conv. quadratically

Higher order is better

think of x_k getting close to x

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Convergence of Newton's Method

some open set/ $r \in (r-\epsilon, r+\epsilon), \epsilon > 0$

Theorem 1: Suppose f, f' , and f'' are continuous in a neighbourhood of a root r of f and that $f'(r) \neq 0$. Then there exists $\delta > 0$ such that if $|r - x_0| \leq \delta$, then all x_k generated by Newton's method satisfy

$$|r - x_k| \leq \delta, \quad |r - x_{k+1}| \leq c(\delta) |r - x_k|^2, \quad \swarrow \text{order of conv. 2.}$$

where

$$c(\delta) = \frac{1}{2} \frac{\max_{|r-x| \leq \delta} |f''(x)|}{\min_{|r-x| \leq \delta} |f'(x)|}. \quad \text{Range } [r-\delta, r+\delta]$$

That is, Newton's method has **quadratic convergence** (i.e., order 2) in some neighbourhood of a root r .

$$e_k = 10^{-6} \Rightarrow e_{k+1} \propto 10^{-12}, \text{ converge very fast}$$

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What is Theorem 1 Saying?

Theorem 1 tells us two important pieces of information:

→ If x_0 is within the basin of conv. of size $\delta > 0$, if we start in here, we don't leave it.

→ We converge quadratically within that basin,

$$e_k \approx 10^{-6} \Rightarrow e_{k+1} \propto |e_k|^2 \approx 10^{-12}$$

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Step 1: Taylor Series Expansion

We begin the proof by writing the Taylor series expansion:

For a root r

$$f(x) = f(x_k) + f'(x_k)(x - x_k) + \frac{1}{2} f''(\xi)(x - x_k)^2 \quad \text{with } \xi \in [x, x_k]$$

$$f(r) = 0 = f(x_k) + \underbrace{f'(x_k)}_{e_k}(r - x_k) + \underbrace{\frac{1}{2} f''(\xi)}_{e_k^2}(r - x_k)^2 \quad \text{or } \xi \in [x_k, x]$$

where ξ is between x_k and r

$$\Rightarrow f(x_k) + f'(x_k)e_k = -\frac{f''(\xi)}{2} e_k^2 \quad (1)$$

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Step 2: Write the Error for x_{k+1}

Using the update rule for Newton's method, the error e_{k+1} for x_{k+1} is

$$\begin{aligned} e_{k+1} = r - x_{k+1} &= r - \left(x_k - \frac{f(x_k)}{f'(x_k)} \right) \\ &= \frac{f(x_k) + e_k f'(x_k)}{f'(x_k)} \quad (2) \end{aligned}$$

$$\text{I + II: } e_{k+1} = -\frac{1}{2} \frac{f''(\xi)}{f'(x_k)} e_k^2, \quad \xi \in [x_k, r]$$

But we don't know ξ

Step 3: Define $c(\delta)$

Recall the coefficient $c(\delta)$ for $\delta > 0$ from the theorem statement:

$$c = \frac{1}{2} \frac{f''(\xi)}{f'(x_k)} \leq \frac{1}{2} \frac{\max_{|x_k - r| < \delta} f''(x)}{\min_{|x_k - r| < \delta} f'(x)} = c(\delta)$$

$$|e_{k+1}| \leq c(\delta) e_k^2$$

$$e_k = r - x_k \leq \delta$$

$$|e_{k+2}| \leq c(\delta) e_k^2 \leq \underbrace{c(\delta) \delta}_p e_k \quad \text{so we've proven quadratic convergence.}$$

choose δ such that $c(\delta)\delta < 1$, ^③ this guarantee that exist
 $\lim_{\delta \rightarrow 0} c(\delta) = \frac{1}{2} \left| \frac{f''(r)}{f'(r)} \right|$ and $f'(r) \neq 0$ by assumption

Step 4: Complete the Proof

Finally, if $|e_k| = |r - x_k| \leq \delta$, then

$$\begin{aligned} |e_{k+1}| &\leq c(\delta) e_k^2 = c(\delta) \cdot |e_k| \cdot |e_k| \\ &\leq c(\delta) \delta |e_k| \\ &= p |e_k| \end{aligned}$$

where $p = c(\delta) \cdot \delta < 1$ using ③

This guarantees $|e_k| = |r - x_k| \leq \delta \quad \forall k$, assuming $e_0 \leq \delta$.

Multiplicity of Roots

We made a key assumption that needs to be addressed: $f'(r) \neq 0$.

If $f'(r) = 0$ and $f''(r) \neq 0$, r is a double root.

E.g. $x=1$ for $f(x)=(x-1)^2$. In general, a root r is of **multiplicity** m if $f^{(k)}(r) = 0$ for all $k = 1 \dots m-1$, and $f^{(m)}(r) \neq 0$. In this case, we can modify Newton's method to
$$x_{k+1} = x_k - m \frac{f(x_k)}{f'(x_k)}$$
 for quadratic convergence.

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The Secant Method

What if we can't actually compute $f'(x)$ for some reason?

we can approximate $f'(x) \approx \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$

so that
$$x_{k+1} = x_k - f(x_k) \left(\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} \right)$$

Secant method converges with order $\phi = \frac{\sqrt{5}+1}{2} \approx 1.62$
golden ratio.

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基于我们之前学的牛顿法，我们想知道：怎么选择initial guess，以及 x_0 会不会影响收敛。

对于一个迭代算法， x_k ：第 k 次迭代得到的近似值， r ：真正的根， $e_k = r - x_k$ ：第 k 步的**误差 (error)**，然后存在一个常数 C ，让 order of convergence d 满足： $e_{k+1} \leq C|e_k|^d$ 。阶数越高，收敛越快。

然后就是我们牛顿法的收敛性分析：

我们假设：

- 函数 $f(x)$ 及其一阶导 $f'(x)$ 、二阶导 $f''(x)$ 都连续；
- r 是方程 $f(x) = 0$ 的真根；
- 并且在这个根处，斜率不为零（即 $f'(r) \neq 0$ ）。

👉 这就保证了函数在根附近是“平滑的”，且不会出现奇怪的拐点或水平线。

三、neighbourhood (邻域)

幻灯片上写的 purple 注释:

some open set $r \in (r - \varepsilon, r + \varepsilon)$

意思是“邻域”:

就是一个小范围, 比如在根 r 附近的一段区间。

$$x \in (r - \delta, r + \delta)$$

在这个范围里, 函数表现“正常”, 不乱跳。

四、主要结论 (Convergence Statement)

在这样的条件下, 存在某个 $\delta > 0$,

只要初始点 x_0 离根不太远 (即 $|r - x_0| \leq \delta$),

那么牛顿法生成的序列 x_k 会满足:

$$|r - x_{k+1}| \leq c(\delta) |r - x_k|^2$$

所以牛顿法的收敛阶是2。

中间的证明说明了, 如果初始点离真根够近, 并且函数够“平滑”, 那么牛顿法一定收敛, 而且收敛得非常快 (quadratically fast)。

之前的证明假设了 $f'(r) \neq 0$, 如果等于零, 那么说明这个zero可能是个重根, 比如 $(x-1)^2$, 牛顿法对于重根可能会变慢或者发散! 为了修复这个问题, 我们额外乘以这个重根的阶数 m 。

Secant method就是当计算导数不可能的时候用差分来模拟导数。差分法的阶数是0.618。